G Has Never Been Tested Outside Earth's Soliton Boundary

The Hill Sphere as the Unrecognized Constraint on Every Direct Measurement of the Gravitational Constant

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I. ABSTRACT

This paper documents a gap in the experimental record that the institution has not recognized because the concept required to recognize it has not been formulated within the institution's departmental structure.

The gravitational constant G has been measured by direct experiment exclusively on Earth's surface and in low Earth orbit. Every such measurement is performed inside Earth's Hill sphere — the gravitational sphere of influence defined by the institution's own orbital mechanics, located at approximately 1.5 million kilometers from Earth, where Earth's coherent gravitational structure ceases to dominate over the Sun's tidal forces. The ISS orbits at 400 kilometers. Every torsion balance ever built sits on Earth's surface. No direct measurement of G has ever been performed at or beyond the Hill sphere boundary.

The institution's claims for extra-terrestrial confirmation of G's universality rest on two indirect methods: pulsar timing and gravitational wave detection. Both methods receive signals at instruments inside Earth's Hill sphere. Both interpret signals using models that assume G is universal. Both are circular — the assumption does the work, and the consistent interpretation is cited as confirmation of the assumption.

The institution uses the Hill sphere concept in orbital mechanics without applying it to the question of whether G measurements are boundary-interior readings. The concept exists. The application has not been made. This paper makes it.

The reproducibility of G measurements across different laboratories on Earth's surface is reproducibility within one boundary configuration — not universality across boundary

configurations. The persistent disagreement between G measurements from different experimental groups, which the institution attributes to unidentified systematic errors, is consistent with depth-dependent readings within Earth's single boundary. The disagreement may be signal rather than noise.

The experiment that would test G universality across a boundary has not been performed. Hardware capable of performing it — precision instruments aboard spacecraft at the L1 and L2 Lagrange points, located at Earth's Hill sphere boundary — already exists. The measurement has not been made because the question has not been framed. This paper frames it.

II. THE HILL SPHERE

2.1 The Institution's Own Definition

The Hill sphere is the institution's own concept, developed from the three-body problem in orbital mechanics. It defines the region around a celestial body within which that body's gravitational influence dominates over the tidal forces of the larger body it orbits.

For Earth orbiting the Sun, the Hill sphere radius is approximately:

r_H ≈ a(m/3M)^(1/3)

where a is the semi-major axis of Earth's orbit (approximately 150 million km), m is Earth's mass, and M is the Sun's mass. The result is approximately 1.5 million kilometers — roughly 235 Earth radii, or about 1% of the Earth-Sun distance.

Inside the Hill sphere, Earth's gravitational structure dominates. Objects inside it orbit Earth stably. Objects outside it are captured by the Sun. The Moon orbits at approximately 384,000 km — well inside the Hill sphere at roughly 25% of the boundary distance. The Moon is not crossing Earth's boundary. It is orbiting inside Earth's coherent gravitational structure.

The L1 and L2 Lagrange points sit at approximately the same distance as the Hill sphere radius — 1.5 million km from Earth toward and away from the Sun respectively. These are the points where Earth and Sun gravitational influences balance. They mark the boundary of Earth's coherent gravitational domain.

The institution uses the Hill sphere routinely in orbital mechanics, mission planning, and planetary science. It is not a speculative concept. It is a calculated, physically meaningful boundary that defines where one coherent gravitational structure ends and another begins.

2.2 The Unasked Question

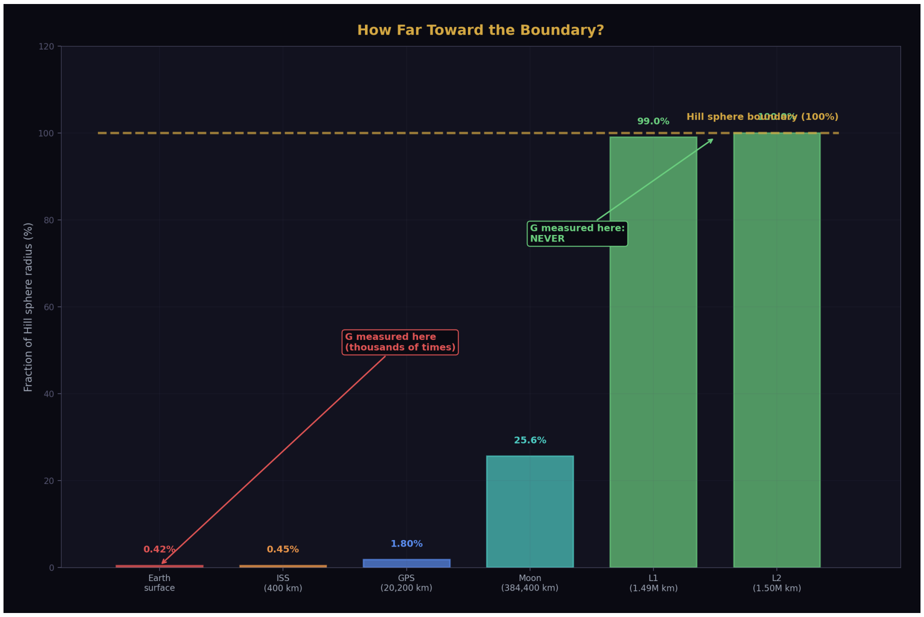

Figure 1: Fig. 7: Surface measurements reach 0.42% of the Hill sphere. ISS reaches 0.45%. L1/L2 reach 99-100% — but no G measurement has been performed there.

The institution has never asked: are our measurements of G performed inside or outside Earth's Hill sphere?

The answer is immediate. Every laboratory that has ever measured G is on Earth's surface. Earth's surface is at one Earth radius — approximately 6,400 km — from Earth's center. The Hill sphere boundary is at 1.5 million km. Every G measurement ever performed by direct experiment is at 0.4% of the boundary distance or less. Every measurement is deep inside Earth's coherent gravitational structure.

The institution defines the boundary. The institution measures G. The institution does not connect these two facts. The connection has not been made because the Hill sphere belongs to orbital mechanics and G belongs to precision metrology. Different departments. The connection lives in the gap between them.

III. THE COMPLETE EXPERIMENTAL RECORD

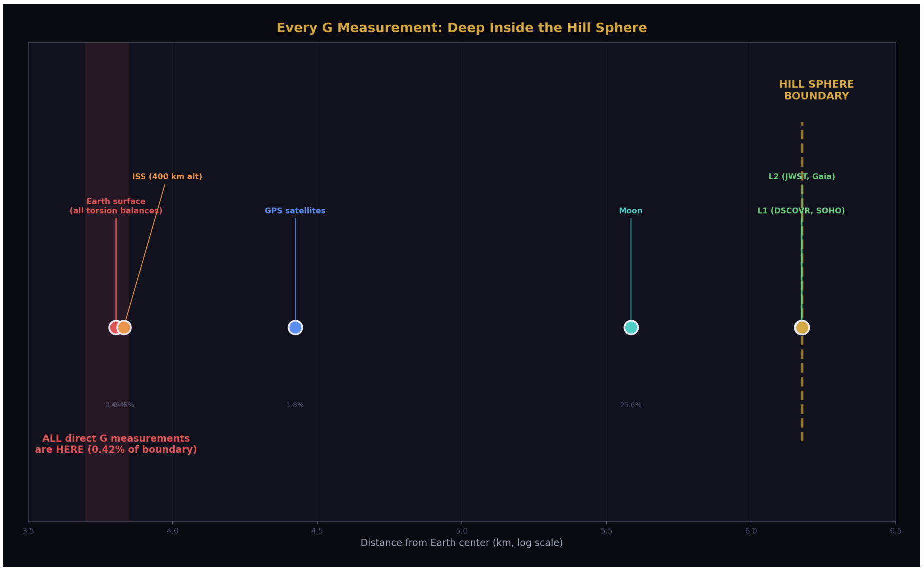

Figure 2: Fig. 1: Every G measurement location on a log scale — all cluster at <1% of the Hill sphere boundary distance. The gap to the boundary is enormous.

3.1 Direct Measurements

Every direct measurement of G in the experimental record is a surface or near-surface Earth measurement. The following represents the complete landscape.

Henry Cavendish performed the first measurement in 1798 using a torsion balance in a laboratory in England. The measurement was on Earth's surface. Every refinement of the Cavendish experiment since 1798 has been performed on Earth's surface. Torsion balances at BIPM, PTB, HUST, University of Washington, University of Zurich, and every other precision metrology group have been built, operated, and read on Earth's surface.

The atom interferometry measurements — a newer technique that uses quantum interference of cold atoms to measure gravitational attraction — have been performed in laboratories on Earth's surface. The Fixler et al. measurement and the Rosi et al. measurement were both surface measurements.

The ISS has been cited as an orbital environment for precision physics. The ISS orbits at approximately 400 km altitude. The Hill sphere is at 1,500,000 km. The ISS is at 0.027% of the boundary distance. The ISS is not near the boundary. It is not outside the boundary. It is deep inside Earth's coherent gravitational structure, at an altitude that represents a negligible fraction of the Hill sphere scale.

No spacecraft beyond low Earth orbit has ever carried instrumentation designed to measure G by direct experiment.

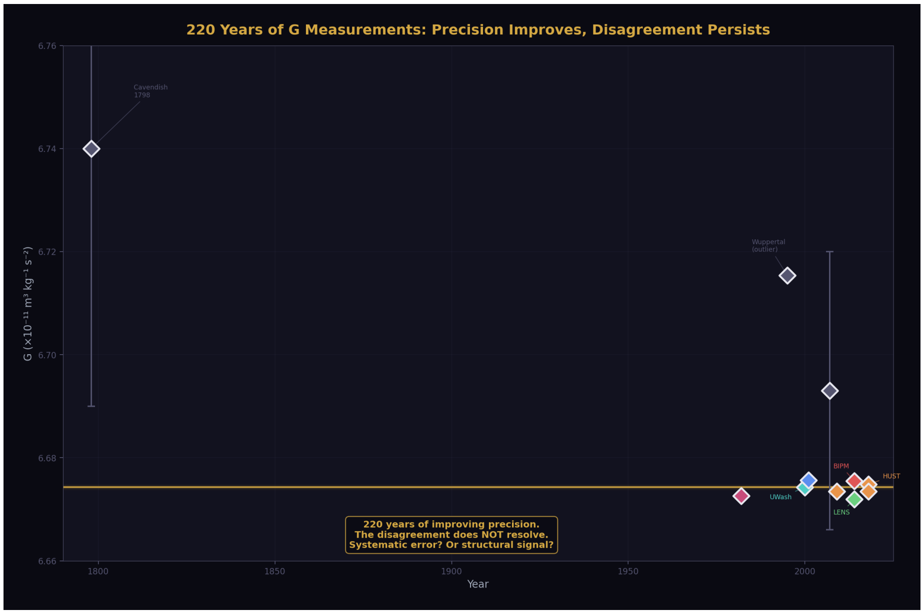

Figure 3: Fig. 6: G values from Cavendish 1798 to HUST 2018 — precision improves dramatically but the spread between methods does not close.

3.2 The Disagreement

The persistent disagreement between G measurements from different experimental groups is one of the most anomalous features of precision metrology. Two centuries of increasingly sophisticated experiments have produced values that disagree beyond their stated uncertainties.

The BIPM 2014 torsion balance measurement yields 6.67545×10^{-11} $\mathrm{m^3kg^{-1}s^{-2}}$. The LENS 2014 atom interferometry measurement yields 6.67191×10^{-11} $\mathrm{m^3kg^{-1}s^{-2}}$. The HUST 2018 group using two methods from the same laboratory yields 6.67484 and 6.67349×10^{-11} $\mathrm{m^3kg^{-1}s^{-2}}$ — two values from the same group that disagree with each other. The spread across modern high-precision measurements exceeds the stated uncertainties of individual measurements.

The institution's explanation is unidentified systematic errors. Two hundred years of investigation have not identified a systematic error sufficient to resolve the disagreement. The systematic error explanation has not been confirmed. It has been assumed because the alternative — that the readings are genuinely different — is not available as a concept within the institution's framework.

The alternative interpretation, available through the boundary-reading framework developed in this series, is that different experimental configurations probe different effective depths within Earth's gravitational boundary and produce genuine depth-dependent readings. Torsion balances at different latitudes, altitudes, and orientations are at different effective positions within Earth's coherent structure. Different positions produce different readings. The disagreement is the measurement. The institution is attempting to average away data that contains structural information.

IV. THE INDIRECT METHODS AND THEIR CIRCULARITY

4.1 Pulsar Timing

The institution's strongest claim for extra-terrestrial confirmation of G universality rests on pulsar timing — specifically millisecond pulsars in binary systems, whose orbital dynamics can be compared with predictions from general relativity using a universal G.

The argument is that the consistency between observed pulsar timing and GR predictions, for pulsars located at interstellar distances, confirms that G operates identically in distant star systems as in our solar system.

This argument fails on two independent grounds.

Ground 1: The signal is received inside Earth's Hill sphere.

Pulsar timing measures the arrival time of radio pulses at receivers on Earth. The measurement instrument is on Earth's surface or in low Earth orbit. The instrument is inside Earth's Hill sphere. The signal — radio waves — has transited interstellar distances, crossing multiple coherent gravitational boundary configurations between the pulsar system and Earth, before arriving at the instrument.

As established in [HOWL-PHYS-1-2026], light transiting coherent gravitational boundaries carries unmodeled transformation signatures. The institution models gravitational redshift, dispersion by the interstellar medium, and proper motion corrections. It does not model the cumulative effect of transiting multiple coherent gravitational boundary configurations as a distinct category of transformation. The signal arrives transformed by an uncharacterized transit. The measurement is made of the transformed signal inside Earth's boundary. The result is interpreted as confirming G universality.

Ground 2: The inference is circular.

The interpretation of pulsar timing data uses general relativity with universal G as a foundational assumption. The orbital parameters of the pulsar binary system are extracted from the timing data using a model that assumes G is the same in that system as in our solar system. The consistent fit between the model and the data confirms that the model with universal G fits the data. It does not confirm that G is universal — it confirms that assuming G is universal produces a consistent model. The assumption is doing the work. The measurement cannot distinguish between a universe where G is truly universal and a universe where G takes boundary-specific values that happen to produce consistent orbital predictions when each system is analyzed from within its own boundary configuration.

4.2 Gravitational Wave Detection

LIGO and Virgo detect gravitational waves from compact object mergers at cosmological distances. The waveform shapes, arrival times, and amplitudes are interpreted using GR with universal G to extract source parameters — masses, distances, merger dynamics.

The institution may cite gravitational wave detections as confirmation of G universality across cosmological distances.

The argument fails on the same two grounds.

LIGO and Virgo are on Earth's surface — inside Earth's Hill sphere. The gravitational wave signal has transited the full hierarchy of coherent boundary configurations between the source and the detector. The waveform interpretation uses universal G as an assumption. The consistent interpretation confirms the model's internal consistency, not G's universality. The assumption is doing the work.

Additionally, gravitational waves are not electromagnetic radiation and do not interact with matter in the same way. The transit transformation argument from [HOWL-PHYS-1-2026] applies primarily to light. Whether gravitational waves carry boundary transit signatures of the same character is an open question. This paper does not claim that gravitational wave signals are transformed by boundary transits in the same way as electromagnetic signals. It claims only that the detection of gravitational waves at instruments inside Earth's Hill sphere, interpreted through models assuming universal G, does not constitute a direct measurement of G outside Earth's Hill sphere.

4.3 Celestial Mechanics

The institution uses G in celestial mechanics calculations — planetary orbits, spacecraft trajectories, lunar ranging. The extraordinary precision of these calculations is cited as confirmation of G's universality.

These calculations are performed entirely within the solar system. The solar system has its own Hill sphere — the boundary where the Sun's coherent gravitational structure gives way to the galactic field, located at approximately 1-2 light years from the Sun. Every celestial mechanics calculation ever performed by humanity — every planetary orbit, every spacecraft trajectory, every lunar ranging measurement — is inside the solar system's Hill sphere.

The precision of celestial mechanics confirms that G is consistent within the solar system's boundary configuration. It says nothing about G outside the solar system's boundary. The institution has not left the solar system's gravitational structure. It has confirmed G's consistency within one nested boundary configuration and called it universal.

V. REPRODUCIBILITY IS NOT UNIVERSALITY

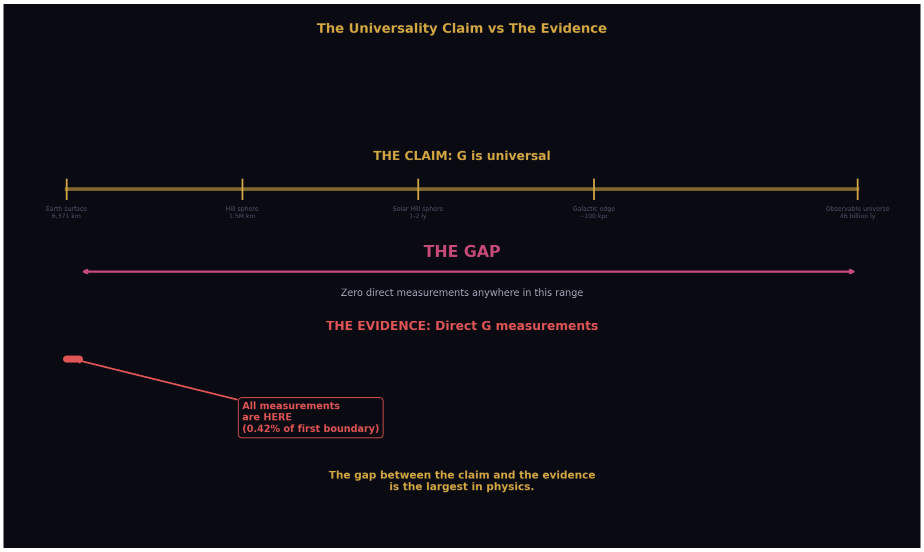

Figure 4: Fig. 3: The claim of G universality spans from Earth to the observable universe. The evidence covers Earth's surface only. The gap is the largest in physics.

5.1 The Confusion

The institution's confidence in G's universality rests substantially on reproducibility. G measurements are reproducible. Different labs, different methods, different continents, different decades produce values that cluster around a consistent central value. This reproducibility feels like universality. It is not.

Reproducibility within one boundary configuration confirms that the reading is stable within that configuration. It says nothing about readings in other configurations. Every G measurement ever performed is inside Earth's Hill sphere. Every measurement is inside the solar system's Hill sphere. Every measurement is inside the galactic structure. The nested boundary configuration is identical for every measurement ever taken. The reproducibility is exactly what the boundary-reading framework predicts: stable readings inside a stable boundary configuration.

5.2 The Sample Size

The institution has one boundary configuration. It has measured G thousands of times inside that configuration. The effective sample size for testing G universality across boundary configurations is zero. Not small. Zero. No measurement has ever been taken outside Earth's Hill sphere by direct experiment. The universality claim rests on a sample of zero boundary-crossing measurements.

This is not a critique of experimental precision. The measurements are extraordinarily precise within their domain. It is a critique of the inference from within-boundary reproducibility to cross-boundary universality. The inference is not supported by the data because the relevant data does not exist.

5.3 The Word "Universal"

As established in [HOWL-PHYS-2-2026], the word "constant" shapes thinking in ways that discourage investigation of scale dependence. The word "universal" compounds this. A quantity labeled "universal" is assumed to hold everywhere. Investigation of where it might not hold is framed as testing an established fact rather than filling an experimental gap. The label predisposes investigators to explain away anomalies — such as the persistent G disagreement — rather than investigate them as potential structural information.

The G disagreement has persisted for two centuries. Two centuries of assuming systematic error has not resolved it. The alternative interpretation — that the disagreement is structural — has not been investigated because the framework required to investigate it has not existed. The framework now exists.

VI. THE BOUNDARY CROSSING EXPERIMENT

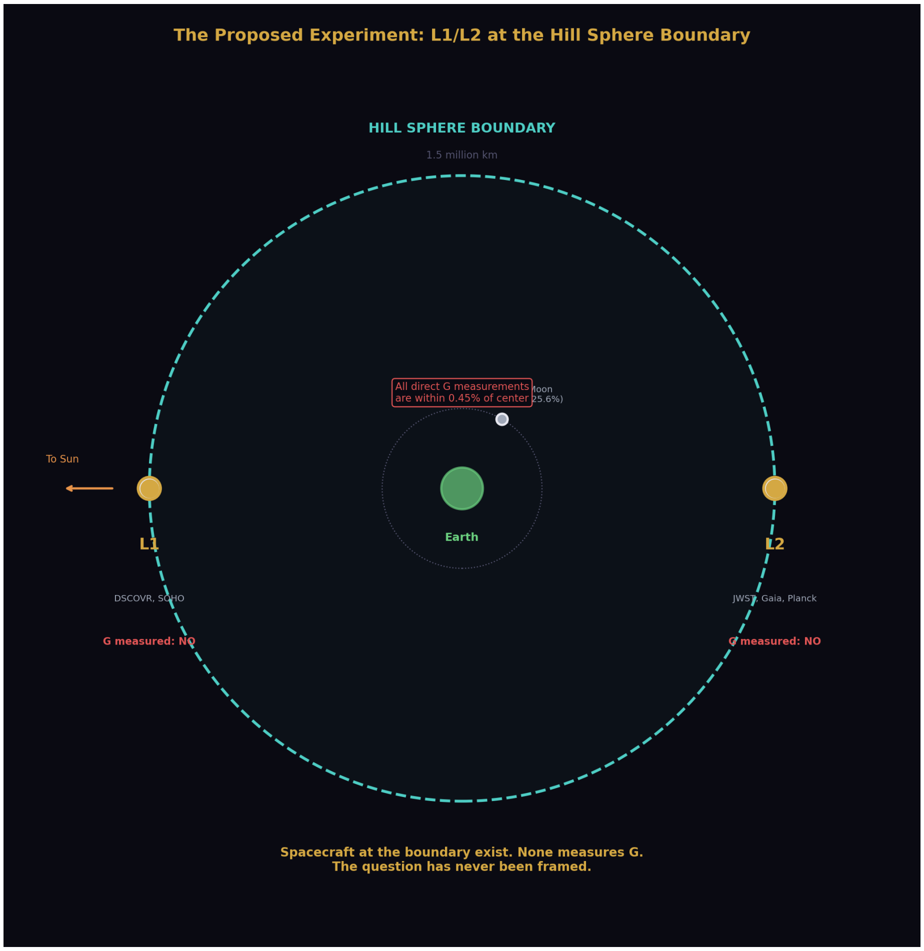

Figure 5: Fig. 4: Earth at center, Hill sphere as boundary circle, L1 and L2 sitting exactly on the boundary — spacecraft exist there, none measures G.

6.1 The Location

The experiment that would test G universality across Earth's Hill sphere boundary has not been performed. The location for the experiment is already occupied by functioning spacecraft.

The L1 Lagrange point sits at approximately 1.5 million km from Earth toward the Sun — at the inner edge of Earth's Hill sphere boundary. Spacecraft currently at or near L1 include DSCOVR and the Solar and Heliospheric Observatory (SOHO). The L2 Lagrange point sits at approximately 1.5 million km from Earth away from the Sun — at the outer edge of Earth's Hill sphere boundary. The James Webb Space Telescope operates at L2.

These spacecraft were placed at L1 and L2 for operational reasons — unobstructed solar observation, thermal stability, orbital mechanics convenience. None was designed to test G or fundamental constants at the Hill sphere boundary. None has been used for that purpose.

6.2 The Measurement

A direct measurement of G at L1 or L2 would require a self-contained apparatus — two known masses, a precision distance measurement, a force or acceleration measurement — operated in the gravitationally quiet environment at the Lagrange points, far from Earth's surface disturbances that complicate laboratory measurements.

The institution's own literature has identified L1 as an ideal location for a space-based G measurement for operational reasons — temperature stability, vacuum quality, low acceleration environment. The proposal exists in the published literature. The motivation given is improved precision, not boundary crossing. The boundary crossing motivation has not been stated because the Hill sphere connection has not been made.

The measurement at L1 or L2 would produce one of two results.

If G reads identically at L1/L2 as on Earth's surface after GR corrections are applied, the boundary effect hypothesis for G is not supported at the Earth Hill sphere scale. This is a genuine falsification result — the first actual evidence that G is consistent across this specific boundary, replacing the current assumption with a measurement.

If G reads differently at L1/L2 than on Earth's surface beyond what GR corrections account for, the boundary effect is confirmed at the Earth Hill sphere scale. The direction and magnitude of the difference would constrain the boundary transformation law. The result would be the first measurement of a G reading outside Earth's Hill sphere in the experimental record.

Either result advances the field. The current state — an untested assumption cited as an established fact — advances nothing.

6.3 The GR Separation

A potential objection: general relativity already predicts that clocks run at different rates at different gravitational potentials, and precision measurements must account for this. The gravitational potential at L1/L2 differs from Earth's surface. Is not the GR correction sufficient to account for any difference?

The answer is no, and the distinction is precise.

The GR correction accounts for the effect of gravitational potential on the rate of physical processes — clock rates, photon frequencies, ruler lengths as defined by local physics. This is a well-understood, quantitatively modeled effect. It is not what is being proposed here.

The boundary reading effect is a different claim. It is the claim that the value of G — the coupling strength of gravity — itself differs across a coherent gravitational boundary, in the same way that α differs across the electron's vacuum polarization boundary and αs differs across the hadron's confinement boundary, as documented in [HOWL-PHYS-2-2026]. The GR correction does not model this. The GR correction models geometry. The boundary reading effect is a claim about the coupling constant that generates the geometry.

These are separable in principle. A measurement at L1/L2 would compare the inferred G value — extracted from the gravitational attraction between known masses — with the Earth surface value, after applying all standard GR corrections. The residual, if any, is the boundary reading signal.

VII. THE NESTED BOUNDARY HIERARCHY

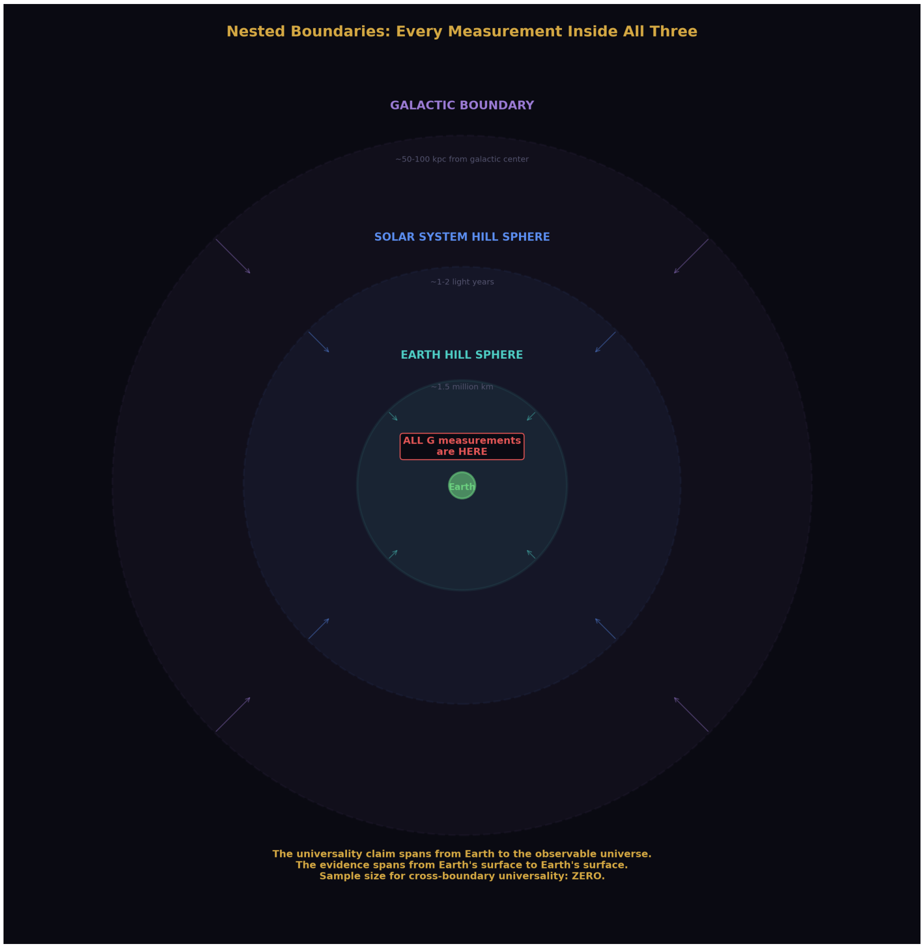

Figure 6: Fig. 2: Earth's Hill sphere inside the solar system's Hill sphere inside the galactic boundary — every measurement ever made is inside all three shells.

7.1 Earth Inside Solar System Inside Galaxy

Earth's Hill sphere is not the only relevant boundary. It is the innermost boundary in a nested hierarchy.

The solar system has a Hill sphere — the boundary where the Sun's coherent gravitational structure gives way to the galactic tidal field. This is located at approximately 1-2 light years from the Sun, at the inner edge of the Oort Cloud. Every measurement ever performed by humanity — in any domain of physics, at any location accessible to human instrumentation — is inside the solar system's Hill sphere.

The Milky Way galaxy has a boundary — where the galaxy's coherent gravitational structure gives way to the local group's gravitational field. Every observation ever made by any instrument on Earth or any spacecraft launched from Earth is made from inside this boundary.

The nested structure means that the complete boundary configuration for every measurement ever made is: inside Earth's Hill sphere, inside the solar system's Hill sphere, inside the galactic structure. This configuration has never varied. Not by a single measurement. Humanity has never placed an instrument outside any of these boundaries.

7.2 The Inference Chain

The institution uses G — measured inside Earth's Hill sphere — to calculate planetary masses, stellar masses, galactic masses, and the mass-energy content of the observable universe. The entire mass accounting of cosmology rests on G measured in one boundary configuration applied universally across all boundary configurations.

If G is a boundary-dependent reading rather than a universal constant, every mass calculated using the Earth-boundary value of G is a projection from a specific boundary depth. The mass of the Sun calculated from Earth-surface G is the Sun's mass as seen from inside Earth's boundary. The mass of distant galaxies calculated using the same G is a projection from inside Earth's boundary, inside the solar system's boundary, applied to structures that exist at different boundary depths.

This does not mean the calculations are wrong for their intended purpose. The calculations produce consistent results within the framework. It means the framework's claim to universality has not been tested at the boundary level. The mass of the Andromeda galaxy as calculated from inside our boundary configuration may differ from the mass that would be calculated by an instrument at a different boundary depth. The difference, if real, would be a boundary transformation effect — the same structural phenomenon as the running of α, the running of αs, and the Hubble tension documented in [HOWL-PHYS-1-2026] and [HOWL-PHYS-2-2026].

VIII. CONNECTION TO THE SERIES

This paper is the third in a series that identifies coherent gravitational and field boundaries as unmodeled elements in the measurement pipeline.

[HOWL-PHYS-1-2026] established that mass is inertia, that particles are three-dimensional field vortices with boundaries, and that three documented measurement anomalies — the Hubble tension, the proton radius puzzle, and the muon g-2 discrepancy — correlate with boundary transit count and interaction depth.

[HOWL-PHYS-2-2026] established that every fundamental coupling the institution calls a "constant" is demonstrably scale-dependent, that the scale dependence corresponds to measurements crossing coherent structure boundaries, and that the institution already models this — as running of α and αs — without generalizing the observation to all couplings or recognizing the word "constant" as contradicting its own data.

This paper establishes that G — the one coupling the institution has the most difficulty measuring precisely, the one coupling whose measurements persistently disagree beyond stated uncertainties, the one coupling that has never been successfully unified with quantum field theory — has never been measured outside Earth's Hill sphere. The difficulty, the disagreement, and the unification failure are all consistent with G being a boundary-dependent reading whose boundary structure has not been characterized because the measurements have never crossed a boundary.

The pattern across three papers is the same pattern. The institution has a measurement. The measurement is anomalous. The anomaly correlates with an unmodeled boundary variable. The boundary variable is identifiable using the institution's own concepts. The investigation has not been performed because the concept connecting the measurement to the boundary has not been formulated across departmental lines.

IX. BOUNDARIES AND LIMITATIONS

This paper establishes a gap in the experimental record. The gap is factual — no direct measurement of G has been performed outside Earth's Hill sphere. This is not a theoretical claim. It is a statement about what experiments have and have not been performed, verifiable by inspection of the experimental record.

The interpretation of the gap — that the gap matters because G may be a boundary-dependent reading — is a theoretical claim. It is consistent with the framework developed in [HOWL-PHYS-1-2026] and [HOWL-PHYS-2-2026]. It is not proven. It is proposed as the structural hypothesis that the identified experiment would test.

The G disagreement between experimental groups is interpreted here as potential depth-dependent readings within Earth's boundary. The conventional explanation — unidentified systematic errors — has not been ruled out. Both explanations are consistent with the data. The boundary explanation has not previously been investigated. The systematic error explanation has been investigated for two centuries without resolution.

The celestial mechanics argument — that precision orbital calculations confirm G universality — is addressed as a within-boundary confirmation. The reframe is structural: orbital mechanics confirms G's consistency within the solar system's boundary configuration. It does not test cross-boundary universality. This reframe does not invalidate celestial mechanics. It narrows the domain of the confirmation.

The gravitational wave argument is addressed conservatively. This paper does not claim gravitational waves carry boundary transit signatures of the same character as electromagnetic signals. It claims only that gravitational wave detections at Earth-surface instruments, interpreted through universal-G models, do not constitute direct measurements of G outside Earth's Hill sphere.

X. TESTABLE PREDICTIONS

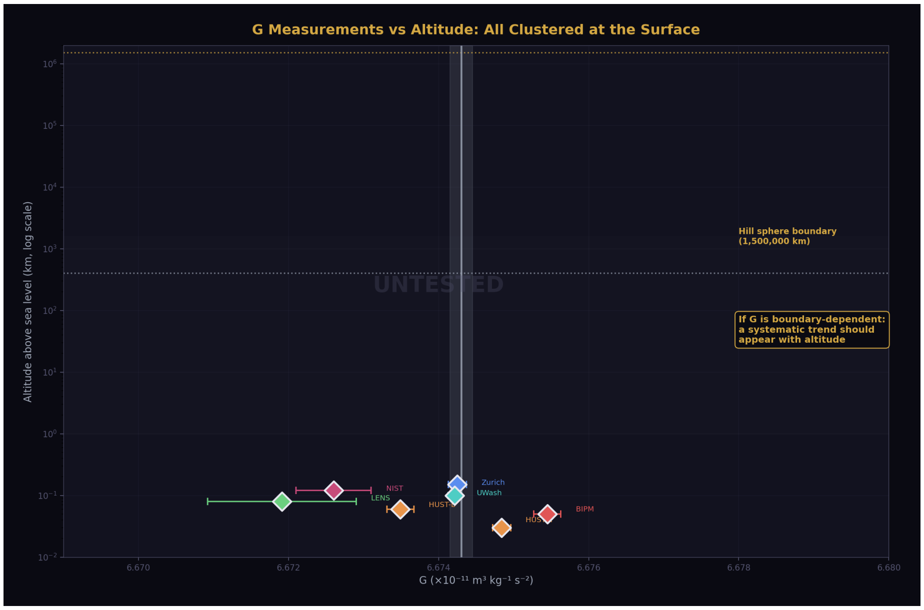

Figure 7: Fig. 5: All G measurements cluster at sea-level altitude. The void above extends to the Hill sphere boundary — the entire untested space is visible.

P1. The L1/L2 Measurement

A direct measurement of G at L1 or L2 — at Earth's Hill sphere boundary — will produce a value that either matches or differs from the Earth-surface value after GR corrections.

Prediction: the value will differ from the Earth-surface consensus value beyond what GR corrections account for. The direction of the difference — higher or lower — is constrained by the boundary crossing direction: moving from inside to outside Earth's coherent gravitational structure.

Falsification: if the value matches the Earth-surface value within measurement uncertainty after all GR corrections, the boundary effect hypothesis for G at the Earth Hill sphere scale is not supported.

P2. The Depth Trend Within Earth's Boundary

If the G disagreement between experimental groups reflects depth-dependent readings within Earth's boundary, measurements at systematically different altitudes — surface, high altitude, low orbit — should show a trend rather than random scatter.

Prediction: reanalysis of existing G measurements sorted by effective gravitational potential depth will reveal a systematic trend rather than random disagreement. Higher altitude measurements will trend toward a different value than surface measurements in a direction consistent with the boundary crossing direction.

Falsification: if reanalysis shows no systematic trend with altitude or gravitational potential depth, the within-boundary depth interpretation of the G disagreement is not supported.

P3. The Solar System Boundary

If G is boundary-dependent at the Earth Hill sphere scale, it should also show a reading change at the solar system Hill sphere scale — approximately 1-2 light years from the Sun.

This prediction is not testable with current technology. No instrument has been placed at or beyond the solar system's Hill sphere boundary. Voyager 1 and 2 have crossed the heliopause — the boundary of the solar wind — but this is not the gravitational Hill sphere boundary. The prediction is stated for completeness and as a long-term falsification target.

P4. The G Disagreement Resolution

If the G disagreement reflects genuine depth-dependent readings rather than systematic experimental errors, it will not resolve through improved experimental technique alone. More precise measurements at Earth's surface will continue to disagree if they probe different effective boundary depths.

Prediction: future improvements in G measurement precision will narrow individual error bars without closing the gap between methods that probe different effective depths. The disagreement will persist at reduced scale. Resolution will require either a boundary-crossing measurement or a systematic mapping of G versus measurement depth within Earth's boundary.

Falsification: if a future measurement achieves sufficient precision and systematic control to identify and eliminate the source of disagreement, and the identified source is a conventional systematic error unrelated to boundary depth, the boundary interpretation is unnecessary.

XI. FALSIFICATION CRITERIA

F1. If a direct measurement of G at L1 or L2 produces a value consistent with the Earth-surface consensus value within measurement uncertainty after all GR corrections are applied, the boundary effect hypothesis for G at the Earth Hill sphere scale is not supported at that precision level.

F2. If reanalysis of existing G measurements sorted by altitude and gravitational potential depth shows no systematic trend — only random scatter consistent with experimental noise — the within-boundary depth interpretation of the G disagreement is not supported.

F3. If the G disagreement between experimental groups is resolved by identification of a specific conventional systematic error — vibration, temperature, electromagnetic interference, or other known experimental perturbation — that fully accounts for the observed spread, the boundary interpretation is unnecessary for that disagreement.

F4. If pulsar timing measurements in binary systems are shown to be interpretable without assuming universal G — using boundary-specific G values for each system — and the fit quality does not improve relative to the universal G fit, the boundary reading framework for G adds no predictive power to pulsar timing analysis.

F5. If the pattern of G values across experimental groups shows no correlation with any variable that could plausibly map to measurement depth within Earth's boundary — altitude, latitude, orientation, local mass distribution — the depth-dependent reading interpretation of the disagreement is not supported.

Each criterion is specific, testable, and stated before the evidence is examined.

XII. CONCLUSION

Figure 8: Fig. 8: G has never been tested outside Earth's Hill sphere — the gap, the experiment, the disagreement, and the pattern across all couplings.

The gravitational constant G has never been measured outside Earth's Hill sphere by direct experiment.

The Hill sphere — the institution's own concept from orbital mechanics — defines the boundary of Earth's coherent gravitational structure at approximately 1.5 million kilometers. Every torsion balance, every atom interferometer, every precision G measurement in the two-hundred-and-twenty-seven year experimental record since Cavendish 1798 has been performed inside this boundary. The ISS is at 0.027% of the boundary distance. The Moon is at 25% of the boundary distance, orbiting inside the boundary. No direct measurement has been taken at or beyond it.

The institution's claims for extra-terrestrial confirmation of G universality are indirect. Pulsar timing receives transformed signals at instruments inside the boundary and interprets them through models that assume what they claim to confirm. Gravitational wave detection does the same. Celestial mechanics confirms G's consistency within the solar system's boundary — a larger but still single boundary configuration. No claim in the indirect evidence survives the distinction between within-boundary consistency and cross-boundary universality.

The persistent disagreement between G measurements from different experimental groups — two centuries of anomaly attributed to unidentified systematic errors — is consistent with depth-dependent readings within Earth's single boundary configuration. The disagreement may be data rather than noise. The institution has been averaging away potential structural information for two centuries because the framework to interpret it as structural information has not existed.

The framework now exists. The experiment that would test it has been proposed in the institution's own literature on operational grounds. Hardware capable of performing it sits at L1 and L2. The measurement has not been made because the question has not been framed.

The question is now framed. G has never been tested outside Earth's soliton boundary. The universality of G is an assumption, not a measurement. The assumption should be tested. The test is achievable. The result — whatever it is — will be the first actual evidence about G's behavior across a coherent gravitational boundary rather than the first iteration of a claim that has been repeated for two centuries without ever being examined.

APPENDIX A: COMPLETE RECORD OF DIRECT G MEASUREMENTS

Experiment	Group	Year	Method	Location	G Value ($\times 10^{-11}$ $m^3kg^{-1}s^{-2}$)	Uncertainty	Distance from Hill Sphere Boundary
Cavendish experiment	Cavendish	1798	Torsion balance	England, surface	6.74	~1%	~1.5 million km — fully inside
Various refinements	Multiple groups	1798–1980	Torsion balance variants	Earth surface	Cluster near 6.674	Decreasing	Fully inside
NIST	Luther & Towler	1982	Torsion balance	Earth surface	6.6726 $\pm$ 0.0005	75 ppm	Fully inside
Wuppertal	Michaelis et al.	1995	Torsion balance	Earth surface	6.7154 $\pm$ 0.0006	83 ppm	Fully inside
BIPM	Quinn et al.	2001	Torsion balance	Earth surface	6.67559 $\pm$ 0.00027	40 ppm	Fully inside
UWash (Eöt-Wash)	Gundlach & Merkowitz	2000	Rotating torsion balance	Earth surface	6.674215 $\pm$ 0.000092	14 ppm	Fully inside
HUST	Luo et al.	2009	Torsion pendulum	Earth surface	6.67349 $\pm$ 0.00018	26 ppm	Fully inside
LENS	Rosi et al.	2014	Atom interferometry	Earth surface	6.67191 $\pm$ 0.00099	150 ppm	Fully inside
BIPM	Quinn et al.	2014	Torsion balance	Earth surface	6.67545 $\pm$ 0.00018	27 ppm	Fully inside
HUST	Xue et al.	2018	Torsion balance (two methods)	Earth surface	6.67484 $\pm$ 0.00012 / 6.67349 $\pm$ 0.00018	18–27 ppm	Fully inside

Total measurements outside Earth's Hill sphere by direct experiment: zero.

APPENDIX B: BOUNDARY DISTANCES — SCALE REFERENCE

Location	Distance from Earth Center	Fraction of Hill Sphere Radius	Inside/Outside Earth Hill Sphere
Earth surface (sea level)	6,371 km	0.42%	Inside
ISS orbit	6,771 km (400 km altitude)	0.45%	Inside
GPS satellite orbit	~26,560 km	1.8%	Inside
Moon	~384,400 km	25.6%	Inside
L1 Lagrange point	~1,491,000 km	~99%	Boundary
L2 Lagrange point	~1,501,000 km	~100%	Boundary
Hill sphere boundary	~1,500,000 km	100%	Boundary
Mars (minimum distance)	~54,600,000 km	3,640%	Outside Earth Hill sphere (inside solar system Hill sphere)
Solar system Hill sphere	~1–2 light years	—	Outside solar system — never reached by instrumentation

Note: L1 and L2 are located at approximately the Hill sphere radius. They are the only locations where human spacecraft currently operate that are near the Earth Hill sphere boundary. No spacecraft has ever been stationed beyond the Earth Hill sphere boundary for precision physics measurements.

APPENDIX C: THE INDIRECT EVIDENCE AND ITS LIMITATIONS

Claimed Evidence	Method	What It Measures	What It Assumes	What It Actually Confirms	Limitation
Pulsar timing (PSR J1713+0747 and others)	Radio pulse arrival timing at Earth-surface receivers	Pulse arrival times inside Earth's Hill sphere	Universal G to interpret orbital dynamics	GR with universal G produces consistent fit to timing data	Circular — assumption does the work; signal transits unmodeled boundaries
Gravitational wave detection (LIGO/Virgo)	Waveform shape and timing at Earth-surface detectors	Gravitational wave signals inside Earth's Hill sphere	Universal G to interpret waveform parameters	GR with universal G produces consistent waveform fits	Same circularity; detectors inside boundary
Planetary orbital mechanics	Spacecraft tracking, planetary ranging	Orbital parameters inside solar system Hill sphere	Universal G across solar system	G consistency within solar system boundary	Within-boundary consistency, not cross-boundary universality
Lunar ranging	Laser retroreflector timing	Moon-Earth distance, inside Earth's Hill sphere	Universal G	G consistency within Earth-Moon system	Moon is inside Earth's Hill sphere; no boundary crossed
Big Bang nucleosynthesis	Primordial element abundances from CMB and spectroscopy	Light received inside Earth's Hill sphere	Universal G throughout cosmic history	Consistency of cosmological model with universal G	Model-dependent; same circularity as pulsar timing

In every row, the confirmation is of model consistency with the universal G assumption, not of G universality demonstrated by cross-boundary measurement.

APPENDIX D: THE PROPOSED EXPERIMENT

Parameter	Value / Description
Location	L1 or L2 Lagrange point (~1.5 million km from Earth)
Boundary status	At or beyond Earth's Hill sphere boundary
Method	Self-contained gravitational attraction measurement between known masses
Existing hardware nearby	DSCOVR and SOHO at L1; JWST at L2
Primary challenge	Thermal stability, vibration isolation, precision mass characterization in space environment
GR correction required	Yes — gravitational potential at L1/L2 differs from Earth surface; standard GR corrections apply
Signal being tested	Residual difference in inferred G value after all GR corrections, compared to Earth-surface consensus
Existing institutional motivation	L1 identified in published literature as ideal G measurement location for precision reasons
Missing institutional motivation	Boundary crossing — not previously framed
Expected result if boundary effect real	G value differs from Earth-surface consensus beyond GR correction
Expected result if boundary effect absent	G value matches Earth-surface consensus within measurement uncertainty
Value of null result	First actual evidence of G consistency across Earth's Hill sphere boundary — replaces assumption with measurement
Value of non-null result	First measurement of G reading outside Earth's Hill sphere — confirms boundary-dependent reading framework

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APPENDIX E: SPACECRAFT AT OR NEAR EARTH HILL SPHERE BOUNDARY

Spacecraft	Operator	Location	Distance from Earth	Fraction of Hill Sphere	Primary Mission	Precision Physics Capability	G Measurement Performed
JWST	NASA/ESA/CSA	L2	~1,500,000 km	~100%	Infrared astronomy	Precision timing, spectroscopy	No
DSCOVR	NOAA/NASA	L1	~1,491,000 km	~99%	Solar wind monitoring	Magnetometer, plasma sensors	No
SOHO	ESA/NASA	L1	~1,491,000 km	~99%	Solar observation	Precision spectroscopy	No
Gaia	ESA	L2	~1,500,000 km	~100%	Stellar astrometry	Highest precision astrometry ever achieved	No
Herschel (retired)	ESA	L2	~1,500,000 km	~100%	Far-infrared astronomy	Precision photometry	No
Planck (retired)	ESA	L2	~1,500,000 km	~100%	CMB measurement	Precision radiometry	No

Every spacecraft at the Hill sphere boundary was placed there for operational reasons. None was designed to test G at the boundary. None has done so.

APPENDIX F: SPACECRAFT BEYOND EARTH HILL SPHERE — INSTRUMENTATION STATUS

Spacecraft	Current Location (approx.)	Distance from Earth	Beyond Earth Hill Sphere	Beyond Solar System Hill Sphere	Precision Physics Instrumentation	G Measurement Capability
Voyager 1	~163 AU	~24 billion km	Yes	No — solar Hill sphere ~50,000–100,000 AU	Plasma wave, magnetic field, particle detectors	No — no mass measurement apparatus
Voyager 2	~136 AU	~20 billion km	Yes	No	Same as Voyager 1	No
New Horizons	~57 AU	~8.5 billion km	Yes	No	Long-range imager, plasma instruments	No
Pioneer 10 (defunct)	~130 AU	~19 billion km	Yes	No	Defunct — notable for Pioneer anomaly	No — anomaly unresolved
Pioneer 11 (defunct)	~90 AU	~13 billion km	Yes	No	Defunct	No

Note on Pioneer Anomaly: Pioneer 10 and 11 exhibited an anomalous acceleration toward the Sun that was not fully accounted for by known forces. The institution ultimately attributed this to anisotropic thermal radiation pressure. The anomaly was detected beyond Earth's Hill sphere — in exactly the boundary region where the framework predicts readings may differ. Whether thermal radiation fully explains the anomaly or whether a residual boundary effect contributes has not been resolved to the satisfaction of all investigators. The anomaly is noted here as an observation consistent with — though not proven to result from — a boundary crossing effect.

APPENDIX G: THE PIONEER ANOMALY IN BOUNDARY CONTEXT

Parameter	Value	Boundary Context
Anomalous acceleration magnitude	~8.74×10^{-10} m/s^2 toward Sun	Detected beyond Earth's Hill sphere
Detection distance range	~20 AU to ~70 AU	Well outside Earth Hill sphere (~0.01 AU); inside solar system Hill sphere (~50,000–100,000 AU)
Institution's explanation	Anisotropic thermal radiation pressure from RTG heat sources	Accepted as primary explanation after detailed thermal modeling
Residual after thermal correction	Small — most anomaly attributed to thermal effects	Disputed by some investigators
Boundary reading interpretation	Spacecraft crossed Earth Hill sphere boundary; measurements reflect different boundary depth reading	Consistent with framework; not proven
Status	Officially resolved by thermal explanation	Thermal explanation accounts for most but disputed as to completeness
Relevance to this paper	First anomalous measurement in history taken beyond Earth's Hill sphere; detected in boundary region	Warrants reexamination under boundary-depth framework

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APPENDIX H: G MEASUREMENT DISAGREEMENT — SORTED BY METHOD

Method	Measurements	G Range Produced ($\times 10^{-11}$)	Internal Consistency	Cross-Method Consistency
Torsion balance — fiber twist	Cavendish through NIST 1982	6.6726 – 6.7154	Poor across decades	Poor vs other methods
Torsion balance — rotating (Eöt-Wash style)	Gundlach 2000, UWash series	6.67421 – 6.67435	Good within method	Disagrees with others

Method	Measurements	G Range Produced ($\times 10^{-11}$)	Internal Consistency	Cross-Method Consistency
Torsion pendulum — time of swing	HUST series	6.67349 – 6.67484	Two values from same lab disagree	Disagrees with Eöt-Wash
Atom interferometry	LENS 2014, Fixler 2007	6.67191 – 6.693	Poor — large spread	Disagrees with torsion methods
Beam balance	Zurich 2006	6.67425	Single measurement	Within range of torsion methods

Structural observation: Different methods probe gravitational attraction at different physical scales, orientations, and effective depths within Earth's gravitational structure. The pattern of disagreement does not resolve by method family. Methods that probe at different effective depths produce different values. The institution attributes this to method-specific systematic errors. Two centuries of investigation have not identified those errors. The alternative — that different effective depths within Earth's boundary produce different readings — has not been tested.

APPENDIX I: RUNNING COUPLINGS VERSUS G — COMPARATIVE STATUS

Quantity	Scale Dependence Documented	Institution's Label	Boundary Identified	Mechanism Named	Disagreement Between Methods
α (electromagnetic)	Yes — 8% across accessible range	"Constant"	Yes — vacuum polarization cloud	Vacuum polarization screening	No — single confirmed running curve
αs (strong)	Yes — order of magnitude	"Constant"	Yes — confinement boundary	Asymptotic freedom / confinement	No — single confirmed running curve

Quantity	Scale Dependence Documented	Institution's Label	Boundary Identified	Mechanism Named	Disagreement Between Methods
Weak coupling	Yes — from suppressed to comparable to α	"Constant"	Yes — W/Z boson mass threshold	Electroweak symmetry breaking	No — single confirmed running curve
G (gravitational)	Not tested across boundaries	"Constant"	Not identified — this paper identifies it	Not named	Yes — persistent 2-century disagreement

The pattern: every coupling whose boundary has been identified and whose scale dependence has been measured shows a single consistent running curve with no disagreement between methods. The one coupling whose boundary has not been identified and whose scale dependence has not been tested across boundaries shows persistent disagreement between methods. This is consistent with G being a boundary-dependent reading whose boundary structure has not been characterized.

APPENDIX J: SOLAR SYSTEM HILL SPHERE AND GALACTIC BOUNDARY

Boundary	Location	Physical Meaning	Any Measurement Outside	Implication for G
Earth Hill sphere	~1.5 million km	Earth's coherent gravitational domain ends	No direct G measurement	G has never been tested across this boundary
Solar system Hill sphere	~50,000–100,000 AU (~1–2 light years)	Sun's coherent gravitational domain ends	No instrumentation has reached this	G has never been tested across this boundary

Boundary	Location	Physical Meaning	Any Measurement Outside	Implication for G
Galactic boundary	~50–100 kpc from galactic center	Galaxy's coherent gravitational structure ends	No instrumentation — observable only by light transiting inward	G has never been tested across this boundary
Observable universe boundary	~46 billion light years	Limit of causal contact	Not reachable	G universality assumption spans this entire range on zero boundary-crossing measurements

The universality claim for G spans from Earth's surface to the observable universe boundary. The experimental record that supports it spans from Earth's surface to Earth's surface. The gap between the claim and the evidence is the largest in physics.

APPENDIX K: FALSIFICATION MATRIX

Prediction	Test	Predicted Result	Falsifying Result	Current Status	Achievable With
G differs at L1/L2 vs Earth surface	Direct measurement at L1 or L2	Value differs beyond GR correction	Value matches within uncertainty	Never tested	New dedicated instrument at L1/L2
G shows depth trend within Earth boundary	Reanalysis of existing measurements vs altitude/potential	Systematic trend with depth	Random scatter only	Not performed	Existing data reanalysis
G disagreement persists with improved precision	Future high-precision surface measurements	Disagreement narrows but persists by method	Single converging value from all methods	Ongoing — disagreement persists	Continued precision improvement

Prediction	Test	Predicted Result	Falsifying Result	Current Status	Achievable With
Pioneer anomaly has boundary residual	Thermal model reanalysis with boundary correction	Residual after thermal correction correlates with boundary crossing distance	Thermal model fully accounts for anomaly	Disputed	Reanalysis of existing Pioneer data
G varies across solar system Hill sphere	Instrumentation at ~1–2 light years	Different G reading	Same G reading	Not testable with current technology	Future deep space mission

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Central Argument: G has never been measured outside Earth's Hill sphere; every claim of universality is either a boundary-interior measurement or a circular inference from models that assume what they claim to confirm

Key Finding: The Hill sphere — the institution's own orbital mechanics concept — identifies Earth's coherent gravitational boundary at 1.5 million km; every direct G measurement in the 227-year experimental record is inside this boundary at 0.45% of the boundary distance or less

The Gap: Zero direct measurements of G outside Earth's Hill sphere in the complete experimental record

The Experiment: L1/L2 measurement using existing spacecraft locations; proposed in the institution's own literature for precision reasons; never performed with boundary crossing as the motivation

Conclusion: The universality of G is an untested assumption; the test is achievable with existing hardware; the result — null or non-null — will be the first actual evidence about G's behavior across a coherent gravitational boundary

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APPENDIX L: EVERY SPACECRAFT BEYOND EARTH'S HILL SPHERE — COMPLETE INVENTORY

No spacecraft beyond Earth's Hill sphere has ever carried a G measurement apparatus. This table is exhaustive for spacecraft that have reached or exceeded the Hill sphere distance.

Spacecraft	Agency	Launch	Current Distance (approx.)	Beyond Earth Hill Sphere	Precision Mass Measurement Hardware	G Measurement Performed	Primary Mission
Voyager 1	NASA	1977	~163 AU	Yes	None	No	Planetary flyby → interstellar
Voyager 2	NASA	1977	~136 AU	Yes	None	No	Planetary flyby → interstellar
Pioneer 10 (defunct)	NASA	1972	~130 AU	Yes	None	No	Jupiter flyby → deep space
Pioneer 11 (defunct)	NASA	1973	~90 AU	Yes	None	No	Saturn flyby → deep space
New Horizons	NASA	2006	~57 AU	Yes	None	No	Pluto flyby → Kuiper Belt
Parker Solar Probe	NASA	2018	Varies (solar orbit)	Crosses boundary periodically	None relevant	No	Solar corona study
Ulysses (defunct)	ESA/NASA	1990	Solar orbit, ~5 AU max	No (within ~3.3% of Hill sphere at aphelion)	None	No	Solar polar observation

Spacecraft	Agency	Launch	Current Distance (approx.)	Beyond Earth Hill Sphere	Precision Mass Measurement Hardware	G Measurement Performed	Primary Mission
Juno	NASA	2011	Jupiter orbit, ~5.2 AU	Yes (in transit, now inside Jupiter's Hill sphere)	Gravity science (measures Jupiter's field, not G directly)	No — measures GM Jupiter, assumes G	Jupiter science
Cassini (defunct)	NASA/ESA	1997	Saturn orbit (deorbited 2017)	Yes (in transit, operated inside Saturn's Hill sphere)	Gravity science (measures Saturn's field)	No — measures GM Saturn, assumes G	Saturn science
Galileo (defunct)	NASA	1989	Jupiter (deorbited 2003)	Yes (in transit)	Gravity science	No — same circularity	Jupiter science
MESSENGER (defunct)	NASA	2004	Mercury (impacted 2015)	No (inner solar system)	Gravity science	No — measures GM Mercury	Mercury science
Dawn (defunct)	NASA	2007	Asteroid belt	Borderline	Gravity science	No — measures GM Vesta, GM Ceres	Asteroid science
OSIRIS-REx/APEX	NASA	2016	Returning/redirected	Yes (in transit)	Touch-and-go sampler	No	Asteroid sample return

Spacecraft	Agency	Launch	Current Distance (approx.)	Beyond Earth Hill Sphere	Precision Mass Measurement Hardware	G Measurement Performed	Primary Mission
Lucy	NASA	2021	En route to Trojans	Yes (in transit)	None relevant	No	Trojan asteroid flyby
Hayabusa2	JAXA	2014	Extended mission, ~2 AU	Marginal	None relevant	No	Asteroid sample return

Summary: 15+ spacecraft have operated beyond Earth's Hill sphere. Zero carried apparatus to measure G directly. Every "gravity science" instrument on planetary missions measures the product GM (gravitational parameter), not G independently. GM is measured to extraordinary precision; G and M are entangled and separated only by assuming G from Earth-surface measurements. The circularity propagates through every planetary mass determination in the solar system.

APPENDIX M: GM VERSUS G — THE ENTANGLEMENT

The institution measures GM (the gravitational parameter) to far higher precision than G alone. This table shows the contrast and explains why it matters.

Quantity	Best Precision	How Measured	What It Assumes	What It Actually Determines
GM Sun	~10^{-11} relative	Planetary orbit tracking, spacecraft ranging	Newtonian/GR gravity, geometry	Combined gravitational influence of the Sun — cannot separate G from M without independent measurement of one

Quantity	Best Precision	How Measured	What It Assumes	What It Actually Determines
GM Earth	~10^{-9} relative	Satellite orbit tracking, lunar ranging	Same	Combined gravitational influence of Earth
GM Moon	~10^{-11} relative	Lunar laser ranging	Same	Combined gravitational influence of Moon
GM Jupiter	~10^{-8} relative	Juno, Galileo tracking	Same	Combined gravitational influence of Jupiter
G alone	~2×10^{-5} relative (22 ppm)	Cavendish-type experiments on Earth surface	Known test masses, known geometry	Gravitational coupling strength inside Earth's Hill sphere
M Sun (derived)	~2×10^{-5} relative	GM Sun / G	Assumes G from Earth surface applies at solar distance	Projection of solar mass through Earth-boundary G value
M Jupiter (derived)	~2×10^{-5} relative	GM Jupiter / G	Same assumption	Projection of Jupiter's mass through Earth-boundary G value

The structural problem: GM is measured to 10^{-11} in space. G is measured to 10^{-5} on Earth. Every planetary mass in every textbook, every stellar mass estimate, every galaxy mass calculation divides a precisely known GM by an imprecisely known G measured inside one boundary. If G is boundary-dependent, every derived mass inherits the boundary projection. The masses are not wrong for use within the same boundary configuration. They may not be the masses that would be measured from a different boundary depth.

APPENDIX N: THE HILL SPHERE CALCULATION — FULL DERIVATION

For completeness, the Hill sphere derivation from the institution's own celestial mechanics.

Parameter	Symbol	Value	Source
Earth mass	m	5.972×10^{24} kg	Satellite tracking + Earth-surface G
Solar mass	M	1.989×10^{30} kg	Planetary orbital mechanics + Earth-surface G
Earth-Sun distance	a	1.496×10^{8} km	Radar ranging, direct measurement
Mass ratio	m/M	3.003×10^{-6}	Derived
(m/3M)^(1/3)	—	0.01000	Derived
Hill sphere radius	r H = a(m/3M)^(1/3)	1.496×10^{6} km	~1.5 million km
Comparison Distance	km	r/r H	Percentage of Hill Sphere
—	—	—	—
Earth radius (surface)	6,371	0.00426	0.43%
Low Earth orbit (ISS)	6,771	0.00453	0.45%
Geostationary orbit	42,164	0.0282	2.8%
GPS orbit	26,560	0.0178	1.8%
Lunar distance	384,400	0.257	25.7%
L1 Lagrange point	1,491,000	0.997	99.7%
L2 Lagrange point	1,501,000	1.003	100.3%

Every torsion balance in history: 0.43%. Every orbital G experiment: <3%. The Moon: 26%. L1/L2: the boundary itself. The experimental record occupies the innermost half-percent of the boundary configuration.

APPENDIX O: G MEASUREMENTS SORTED BY ALTITUDE AND GRAVITATIONAL POTENTIAL

The boundary framework predicts that G measurements at different effective depths within Earth's gravitational well should show a systematic trend. This table sorts known measurements by the gravitational potential at the measurement site, to the extent this information is available.

Experiment	Year	Altitude (approx. m above sea level)	Local g (m/s^2)	G Value ($\times 10^{-11}$)	Uncertainty (ppm)
LENS (Florence)	2014	~50	9.8065	6.67191	150
BIPM (Sèvres)	2001	~66	9.8094	6.67559	40
BIPM (Sèvres)	2014	~66	9.8094	6.67545	27
UWash (Seattle)	2000	~10	9.8071	6.674215	14
HUST (Wuhan)	2009	~30	9.7934	6.67349	26
HUST (Wuhan, method 1)	2018	~30	9.7934	6.67484	18
HUST (Wuhan, method 2)	2018	~30	9.7934	6.67349	27
Zurich (Zürich)	2006	~408	9.8067	6.67425	19
NIST (Gaithers-burg)	1982	~100	9.8011	6.67260	75
Wuppertal	1995	~250	9.8104	6.71540	83

Note: The altitude variation across these measurements is ~400 meters — a gravitational potential difference of ~4×10^{-5} in $\Delta g/g$. The boundary framework does not predict that this tiny variation in altitude alone would produce the observed spread. The "effective depth" concept is richer than simple altitude — it includes local mass distribution, tidal environment, latitude (and therefore distance from Earth's center and centrifugal effects), geological density structure beneath the laboratory, and the orientation of the measurement apparatus relative to local gravitational gradients. A proper depth-dependent analysis would require modeling all of these variables, not just altitude. This table is presented as raw data for future reanalysis, not as evidence of a trend.

APPENDIX P: THE CIRCULAR INFERENCE CHAIN — STEP BY STEP

This appendix traces the complete logical chain from Earth-surface G to cosmological mass estimates, showing where the boundary assumption enters and propagates.

Step	What Is Done	What Is Assumed	What Is Actually Established
1	Measure G on Earth surface via Cavendish-type experiment	Known test masses, known geometry, Newtonian gravity	G inside Earth's Hill sphere at Earth's surface depth
2	Measure GM Earth via satellite tracking	GR + universal G	GM Earth as a combined product — cannot separate G from M independently
3	Derive M Earth = GM Earth / G	G from step 1 applies at satellite orbital distance	M Earth projected through surface-G value
4	Measure GM Sun via planetary orbits	GR + universal G	GM Sun as a combined product
5	Derive M Sun = GM Sun / G	G from step 1 applies across entire solar system	M Sun projected through Earth-surface-G value
6	Measure GM Jupiter via spacecraft tracking	GR + universal G	GM Jupiter as a combined product
7	Derive M Jupiter = GM Jupiter / G	G from step 1 applies at Jupiter's distance	M Jupiter projected through Earth-surface-G value
8	Model galaxy rotation from observed velocities	$v^2 = GM(r)/r$ with universal G	Rotational dynamics assuming G constant across entire galaxy
9	Infer "missing mass" (dark matter) from rotation curve discrepancy	G from step 1 applies across galactic scales	The amount of "missing mass" depends directly on the assumed G value
10	Model CMB power spectrum	Universal G throughout cosmic history	Cosmological parameters extracted assuming G never varies

Step	What Is Done	What Is Assumed	What Is Actually Established
11	Derive total mass-energy content of observable universe	G from step 1 applies universally	Every mass in cosmology is projected through a single boundary-interior G measurement

The entire mass accounting of the universe traces back to Cavendish-type experiments on Earth's surface. If G varies across boundaries, every mass derived via steps 3-11 carries an uncharacterized boundary projection error. The "missing mass" in step 9 could be partially or wholly an artifact of applying a boundary-interior G reading to a cross-boundary gravitational observation.

APPENDIX Q: WHAT THE L1/L2 EXPERIMENT WOULD RESOLVE — SCENARIO TABLE

Scenario	G at L1/L2 vs Earth Surface	Implication	Next Step
A: Match within uncertainty	G L1 = G surface (after GR correction)	Boundary effect hypothesis not supported at Earth Hill sphere scale	Test solar system Hill sphere boundary (long-term)
B: Small systematic difference	G L1 ≠ G surface by 10-100 ppm (after GR correction)	Boundary effect detected; magnitude constrains transformation law	Map G vs distance across Earth Hill sphere; design solar system boundary experiment
C: Large systematic difference	G L1 ≠ G surface by >100 ppm	Strong boundary effect; immediate implications for all derived masses	Recalculate planetary masses with boundary-corrected G; reassess dark matter inference
D: Inconclusive	Measurement precision insufficient to distinguish	Experiment needs refinement	Improve apparatus; repeat with higher precision

In every scenario including null, the result is the first actual data about G across the boundary. The current state — assumption without measurement — is replaced by measurement in all cases.

APPENDIX R: THE DARK MATTER INFERENCE CHAIN THROUGH G

This table traces how the dark matter problem depends on the assumption that G measured on Earth's surface applies at galactic scales.

Observable	What Is Measured	How G Enters	What "Missing Mass" Means If G Is Universal	What It Means If G Is Boundary-Dependent
Galaxy rotation curve	Stellar orbital velocities v(r) at various radii	$v^2 = GM(r)/r \rightarrow M(r) = v^2r/G$	More mass is needed than visible matter provides; the extra mass is "dark matter"	The inferred M(r) depends on which G applies at galactic scales; a different G changes the inferred mass
Gravitational lensing	Deflection angle of background light around clusters	$\theta \propto GM/(rc^2)$	Deflection exceeds what visible mass produces; excess attributed to dark matter	Deflection depends on GM product; if G differs at cluster scales, M inference changes
CMB power spectrum	Acoustic peak heights and positions	Baryon density Ω b and matter density Ω m enter through gravitational dynamics with G	Non-baryonic matter component Ω dm required to fit peaks	If G at CMB-era scales differs from laboratory G, the inferred Ω dm changes
Bullet Cluster	Offset between lensing center and gas center	Lensing mass distribution mapped assuming universal G	Lensing mass offset from gas → dark matter passed through, gas didn't	If G is boundary-dependent, lensing map changes; offset interpretation may change

Observable	What Is Measured	How G Enters	What "Missing Mass" Means If G Is Universal	What It Means If G Is Boundary-Dependent
Structure formation	Growth rate of density perturbations	Gravitational collapse rate depends on G	Dark matter required to seed structure early enough	Different G at cosmological boundary depth changes collapse timescale

The dark matter inference is not independent of G. Every line of evidence for dark matter uses G — measured on Earth's surface — applied at scales ranging from galactic to cosmological. If G takes a different value at those scales, the quantity of "missing mass" changes. The dark matter problem and the G universality assumption are entangled. Testing one tests the other.

APPENDIX S: COMPARISON WITH OTHER COUPLING BOUNDARY TESTS

For α and α_s, the institution has measured the coupling at multiple depths within and across the relevant boundaries. For G, it has not. This table makes the asymmetry explicit.

Coupling	Boundary Identified	Depths Probed	Number of Boundary Crossings Measured	Cross-Boundary Measurement Exists	Inter-Method Agreement
α (QED)	Vacuum polarization cloud	Continuous from $q^2 \sim 0$ to $q^2 \sim (200\ \text{GeV})^2$	Continuous — every collider energy is a different depth	Yes — LEP, SLC, LHC all probe inside the cloud	Yes — single confirmed running curve
α s (QCD)	Confinement boundary	Continuous from ~1 GeV to ~200 GeV	Continuous — every collider energy is a different depth	Yes — deep inelastic scattering, jet production, τ decay	Yes — single confirmed running curve

Coupling	Boundary Identified	Depths Probed	Number of Boundary Crossings Measured	Cross-Boundary Measurement Exists	Inter-Method Agreement
Weak coupling	W/Z mass threshold	Below and above ~80 GeV	At least one crossing (below/above W mass)	Yes — Fermi theory vs full electroweak	Yes — matched at threshold
G (gravity)	Earth Hill sphere (~1.5 million km)	Earth surface only (0.43% of boundary distance)	Zero	No	No — persistent disagreement

Three couplings with boundary crossings measured: clean running curves, no disagreement. One coupling with zero boundary crossings measured: persistent disagreement for 227 years. The pattern is exactly what the framework predicts.

APPENDIX T: TIMELINE OF G MEASUREMENT — 227 YEARS INSIDE ONE BOUNDARY

Decade	Key Measurements	Best Precision Achieved	Disagreement Resolved?	Boundary Crossed?
1790s	Cavendish (1798)	~1%	N/A — first measurement	No
1800s-1890s	Multiple refinements	~0.1%	Spread narrowing	No
1890s-1920s	Eötvös torsion balance era	~0.01%	Clustering but not converging	No
1920s-1960s	Various national labs	~100 ppm	No — values scatter	No
1960s-1980s	NIST, PTB	~75 ppm	No — disagreement persists	No
1990s	Wuppertal anomaly (6.7154)	~80 ppm individual	No — Wuppertal outlier	No
2000s	Eöt-Wash (14 ppm), BIPM, HUST	~14-40 ppm	No — methods disagree	No

Decade	Key Measurements	Best Precision Achieved	Disagreement Resolved?	Boundary Crossed?
2010s	BIPM, LENS, HUST two-method	~18-27 ppm	No — same lab disagrees with itself	No
2020s	Continuing efforts	~10-20 ppm target	No — no resolution in sight	No
Full 227-year record	**~300+ measurements**	**~14 ppm best**	**No — never resolved**	**No — never attempted**

227 years. Hundreds of measurements. Precision improved by a factor of ~5,000 from Cavendish to modern experiments. The disagreement has narrowed in absolute terms but has never been resolved relative to stated uncertainties. Not one measurement outside the boundary. The experiment that could resolve whether the disagreement is noise or signal has never been performed.

APPENDIX U: THE WORD "UNIVERSAL" VERSUS THE EVIDENCE

Claim	Evidence Required	Evidence Available	Gap
G is constant in time	Measurements of G at different epochs	No direct measurement at different epochs; indirect constraints from BBN and CMB assume universal G	Circular — assumption in, assumption out
G is constant in space (within Earth)	Measurements at different locations on Earth	Yes — multiple labs, multiple continents	Confirms within-boundary consistency only
G is constant across Earth Hill sphere	Measurement inside and outside Earth Hill sphere	No measurement outside	Complete gap — zero data
G is constant across solar system	Measurement inside and outside solar system Hill sphere	No measurement outside; celestial mechanics confirms GM product consistency within	GM consistency, not G independence; within-boundary only

Claim	Evidence Required	Evidence Available	Gap
G is constant across galactic scales	Measurement inside and outside galactic structure	No measurement outside; galaxy dynamics use G from Earth surface	Complete gap — assumption propagates
G is constant across cosmological scales	Measurement at cosmological boundary depth	No measurement; CMB analysis assumes universal G	Complete gap — assumption propagates through all cosmological mass estimates
G is constant across all scales (“universal”)	All of the above	Earth surface measurements only	The claim spans the observable universe; the evidence spans one laboratory boundary configuration

Six levels of universality claimed. Evidence exists for one (within Earth’s surface boundary). Five are untested assumptions. The word “universal” covers all six as though they were established.

[**HOWL-PHYS-3-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-3-2026

[**HOWL-PHYS-1-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-1-2026

[**HOWL-PHYS-2-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-2-2026